

Intelligent Placement Prediction System Using Student Academic Data

1. Introduction

An intelligent placement prediction system predicts whether a student is likely to be placed based on academic performance, aptitude, communication skills, internships, and technical skills. Machine learning helps institutions identify students who need additional training.

2. Architecture Diagram / Flowchart

Student Dataset → Data Preprocessing → Feature Selection → Train/Test Split → Machine Learning Model → Placement Prediction → Result

3. Implementation

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Jupyter Notebook

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

df = pd.read_csv('placement_data.csv')
X = df.drop('Placement', axis=1)
y = df['Placement']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
```

4. Results

Explanation of Results

The Random Forest model predicts whether students are likely to be placed. Higher accuracy indicates better prediction performance and helps colleges identify students requiring additional support.

Link of Runnable Project

https://colab.research.google.com/drive/1FFub_l2pQNYL_D7oIV5Wn2GfQX--Efr#scrollTo=ouT52CeG0wzQ

Github Link: <https://github.com/ommalewar8154-wq/Intelligent-Placement-Prediction-System>

5. Viva Questions (One-line Answers)

Q. What is placement prediction?

A. Predicting whether a student will be placed.

Q. Which algorithm is used?

A. Random Forest Classifier.

Q. What is the target variable?

A. Placement status.

Q. Why split the dataset?

A. To train and test the model.

Q. Main benefit?

A. Improves placement preparation.

Output Screenshots

... First 5 Records

	CGPA	Aptitude	Communication	Internship	Placement
0	8.5	80	75	1	1
1	7.2	65	60	0	0
2	9.1	90	85	1	1
3	6.8	55	50	0	0
4	8.0	78	72	1	1

Accuracy: 100.0 %

Confusion Matrix

```
[[2 0]
 [0 2]]
```

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2
1	1.00	1.00	1.00	2
accuracy			1.00	4
macro avg	1.00	1.00	1.00	4
weighted avg	1.00	1.00	1.00	4

Figure 1: Model accuracy, confusion matrix, and classification report.

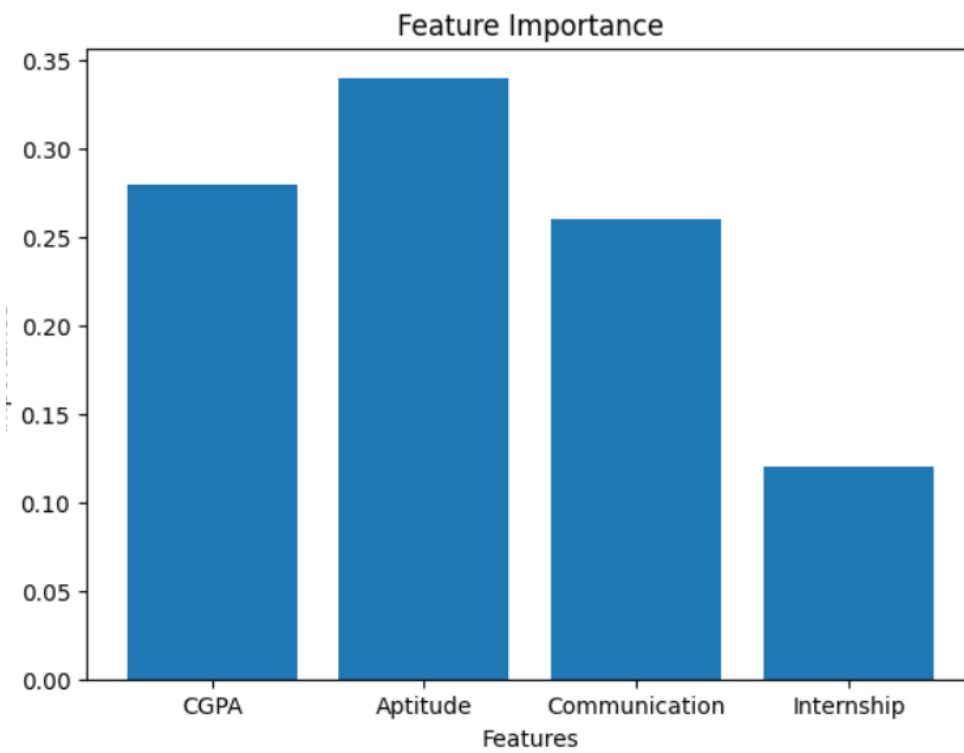


Figure 2: Feature importance graph.